

Swiggy Food Orders

Data Analytics & Visualization

GitHub:

https://github.com/Shreyashgol/SectionD_G4_swiggy

Tableau:

https://public.tableau.com/app/profile/shashwat.sinha3668/viz/Swiggy_Restaurants_Analytics/ExecutiveOverview?publish=yes

FOOD TECH

Team ID

SectionD _ Group 4

Team Members

- 1 . Shreyash Golhani
- 2 . Shashwant Sinha
- 3 . Saksham Sontakke
- 4 . Kavya Mukhija
- 5 . Yash
- 6 . Shivam Dixit

Faculty Mentor

Satyaki Das

Context & Problem Statement

Sector Context

India's online food delivery market exceeded ₹500 billion in 2024, with Swiggy serving 500+ cities. Restaurant owners, category managers, and city-level operations heads need data-driven insights to optimise menus, pricing, and service quality.

Problem Statement

Which cities, categories, and price segments drive the most orders and highest ratings on Swiggy — and what patterns in time and location reveal unmet demand?

Objective

Deliver a clean, analysis-ready dataset and interactive Tableau dashboards that enable business stakeholders to identify top-performing restaurants, optimise pricing tiers, and uncover temporal demand patterns across 28 Indian cities.

Data Engineering

Source → ETL Pipeline → Clean Dataset

Dataset Source

Source: Kaggle

File: swiggy_data_raw.csv

Size: 1,97,430 rows × 10 columns

Period: Jan 2025 – Aug 2025

Python ETL — Key Steps

01_extraction.ipynb

Load & profile raw CSV — shape, nulls, dtypes

02_cleaning.ipynb

Drop 9,871 nulls · Remove 14 duplicates · Filter 75,133 zero-rated rows · IQR outlier flag (4,180 rows >₹577)

02_cleaning.ipynb — Key transformations

```
df = df.dropna()
df = df.drop_duplicates()
df = df[df['Rating Count'] != 0]
df['is_price_outlier'] = ~df['Price (INR)'].between(lower, upper)
```

Key Final Columns

- ▶ state / city / location
- ▶ order_date → month, day_of_week
- ▶ price_inr → price_bucket
- ▶ rating / rating_count
- ▶ category / dish_name

Extract

Clean

EDA

Stats

Final Load

KPI & Metrics Framework

Total Orders

91,788

Why: Measures overall platform activity and dataset coverage

How: `COUNT(dish_name)`

→ `03_eda.ipynb`

Total Revenue

₹2.30 Cr

Why: Reflects cumulative monetary value of all listed dishes

How: `SUM(price_inr)`

→ `03_eda.ipynb`

Avg Price (INR)

₹250.24

Why: Benchmarks pricing strategy across cities and categories

How: `AVG(price_inr)`

→ `04_statistical_analysis.ipynb`

Avg Rating

4.31 / 5

Why: Tracks customer satisfaction across all restaurants

How: `AVG(rating)`

→ `04_statistical_analysis.ipynb`

Total Restaurants

898

Why: Scope of restaurant network across 28 cities

How: `COUNTD(restaurant_name)`

→ `03_eda.ipynb`

Peak Day

Saturday

Why: Weekends prompts for higher customer acquisition

How: `MAX(COUNT by day_of_week)`

→ `05_final_load_prep.ipynb`

Key Insights

Exploratory Data Analysis · 03_eda.ipynb

01

Bengaluru Dominates Order Volume

Bengaluru accounts for 10,385 orders — 1.9× more than the next city (Mumbai: 5,459), making it the single largest market on the platform.

02

Mid-Range Pricing Wins the Market

59.1% of all dishes are priced ₹150–₹400 (Mid-range). Only 11.6% fall in the Premium tier (>₹400), indicating strong price sensitivity among Swiggy users.

03

Customer Satisfaction is Consistently High

The average rating across 898 restaurants is 4.31/5.0, with a right-skewed distribution — suggesting most restaurants maintain strong quality standards.

04

Weekend Demand Peaks on Saturdays

Saturday drives the highest order volume (13,538 orders), followed by Thursday and Sunday — revealing a clear mid-week and weekend demand surge.

05

Orders are Stable Across Jan–Aug 2025

Monthly orders remain consistently between 10,876–11,882, with a slight uptick in Aug 2025 — indicating steady platform growth with no seasonal crash.

06

40% of Raw Data Flagged as Unreliable

75,133 rows had Rating Count = 0 and 9,871 rows had missing values — nearly 43% of raw data was filtered before analysis, highlighting significant data quality gaps.

Advanced Analysis

Statistical Testing · 04_statistical_analysis.ipynb · scipy library

ANOVA	Does Rating differ significantly across Cities? F = 62.97 p < 0.0001 ✓ Significant scipy.stats.f_oneway() City location has a strong statistically significant effect on restaurant ratings. Bengaluru and Chennai restaurants consistently outperform other cities — city-specific quality standards matter.
ANOVA	Does Rating differ significantly across Categories? F = 3.12 p < 0.0001 ✓ Significant scipy.stats.f_oneway() Food category influences customer satisfaction. Certain categories consistently attract higher ratings — suggesting menu specialisation drives better customer experience.
Spearman	Is there a significant correlation between Price & Rating? ρ = 0.029 p < 0.0001 ✓ Significant but Weak scipy.stats.spearmanr() Price and Rating are statistically linked but practically unrelated (ρ = 0.029). Higher price does NOT guarantee higher ratings — quality perception is independent of price on Swiggy.

Tableau Dashboard Walkthrough

3 Interactive Dashboards · Fixed 1200×800 · Swiggy Brand Theme · Tableau Public

D1

Overview Analytics

Executive Overview

KPIs: Orders · Revenue · Rating · Average Price

Charts: Top 5 Cities by Orders (Bar) · Orders by Price Bucket (Pie)

D2

Restaurants & Price Analytics

Optimized Pricing Strategy | Indentify Winners

KPIs: Total Orders · Total Revenue · Avg Price · % Premium Orders

Charts: Price vs Rating Scatter · Segment Info Cards · Restaurants Table

D3

Temporal Patterns & Customer Segments

Optimized Staffing | Target Marketing

KPIs: Peak Day · Lowest Weekday · Monthly Trend

Charts: Days of Week · Month Trend · Top Categories · Segment Strategy Cards

Common Slicers (all dashboards): State · City · Category · Price Bucket · Order Date Range

Tableau URL: [Swiggy Restaurants Analytics](#)

Recommendations

R1	<i>Insight: Bengaluru generates 1.9× more orders than any other city</i> → Launch city-specific restaurant partnerships and exclusive Bengaluru deals to deepen market leadership in the top city	Impact Estimated 10–15% revenue uplift from top market
R2	<i>Insight: 59.1% of orders are Mid-range (₹150–₹400); Premium only 11.6%</i> → Introduce 'Premium Upgrade' nudges at checkout and curate a Premium restaurant collection to shift 5% of Mid-range users to Premium	Impact ~₹11.5L incremental revenue from tier migration
R3	<i>Insight: Price and Rating are statistically unrelated ($p = 0.029$)</i> → Market budget restaurants with high ratings more aggressively — value-for-money positioning is a proven driver of orders independent of price	Impact Higher conversion among price-sensitive users
R4	<i>Insight: Saturday & Thursday drive peak demand; Monday is lowest</i> → Launch weekday promotions (Monday–Tuesday discounts) and scale delivery capacity on Saturdays to reduce drop-offs during peak hours	Impact 15–20% reduction in order cancellations on peak days
R5	<i>Insight: 43% of raw data had quality issues (nulls + zero ratings)</i> → Implement mandatory restaurant data completion checks before listing — enforce minimum rating count thresholds for visibility on the platform	Impact Cleaner data → more reliable recommendations

Impact & Value

The "So What?" — Why stakeholders must act on these findings

₹2.30 Cr

Total Revenue
Analysed

43%

Raw Data Cleaned
Before Analysis

4.31★

Platform Avg
Rating

28

Cities &
States Covered

Cost of Inaction

Without data-driven decisions, Swiggy risks losing high-value markets like Bengaluru to competitors who invest in localised restaurant partnerships and targeted pricing.

Revenue Opportunity

Shifting 5% of Mid-range users to Premium tier across 91,788 orders could unlock ~₹11.5L in incremental revenue — achievable through checkout nudges alone.

Operational Efficiency

Scaling delivery capacity on peak days (Saturday +13.5K orders) and reducing Monday idle time can cut operational costs by an estimated 12–18% through demand-matched staffing.

Data Quality ROI

Enforcing minimum data standards at listing could recover 43% of currently unusable data — directly improving recommendation accuracy and customer experience scores.

Limitations & Next Steps

Limitations

Dataset is synthetic — findings may not reflect real Swiggy platform behaviour
43% of raw data dropped — analysis based on 91,788 of 1,97,430 original rows
No customer-level data — cannot analyse repeat orders or churn behaviour
No delivery time or complaint data — service quality analysis is incomplete
Order Date range limited to Jan–Aug 2025 — no seasonal trend beyond 8 months

Next Steps

01	Build a predictive model for restaurant rating using price, category, and city features (sklearn)
02	Integrate real-time Swiggy API data for live dashboard updates in Tableau
03	Extend dataset to full year (Jan–Dec) for complete seasonal trend analysis
04	Add customer segmentation (RFM model) once order-level customer data is available
05	Deploy ETL pipeline as a scheduled job (Apache Airflow) for automated weekly refresh

Team & Contribution

Section D · Group 4 · DVA Capstone 2

Data Engineering

01_extraction · 02_cleaning

Lead: Shivam Dixit

EDA & Visualization

03_eda

Lead: Kavya mukhija

Statistical & Business Analysis

04_statistical_analysis

Lead: Shreyash Golhani

KPI'S Building

Lead: Saksham Sontakke

Tableau Dashboards

Lead: Shashwat Sinha

Report & PPT Making

Lead: Yash

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